Sales Forecasting System

1. Introduction

The Sales Forecasting System is a data analysis and machine learning project developed to analyse historical sales data and predict future sales. The project uses Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn to perform data preprocessing, trend analysis, visualization, and sales forecasting.

Historical sales data is analysed on a monthly and yearly basis to understand sales patterns and trends. A Linear Regression model is then trained using a time-based feature to forecast future sales.

2. Objective

The main objectives of this project are:

- To clean and preprocess historical sales data.
- To analyse monthly and yearly sales trends.
- To visualize historical sales patterns.
- To build a sales forecasting model using Linear Regression.
- To evaluate the forecasting model using MAE, MSE, and RMSE.
- To predict sales for the next six months.
- To generate business insights that can support sales planning and decision-making.

3. Dataset Description

The dataset contains historical sales transaction records with information about orders, customers, products, locations, and sales.

The `Order Date` column was used as the primary time variable, while the `Sales` column was used as the target variable for sales analysis and forecasting.

Important fields used in the project include:

- `Order Date` — date of the customer order.
- `Ship Date` — date when the order was shipped.
- `Category` — product category.
- `Sub-Category` — product sub-category.
- `Region` — geographical sales region.
- `State` — customer state.
- `City` — customer city.
- `Sales` — sales amount.

The dataset does not contain a Quantity column, so quantity-based analysis was not performed.

4. Data Cleaning and Preprocessing

The dataset was inspected for missing values, duplicate records, incorrect data types, and basic data quality issues.

There were 11 missing values in the `Postal Code` column. All missing records belonged to Burlington, Vermont. The missing values were filled with the corresponding Burlington, Vermont postal code, represented as `54001` in the numeric column.

The `Order Date` and `Ship Date` columns were converted from string format to datetime format to enable time-based analysis.

Additional time-based features were created from `Order Date`, including:

- Order_Year
- Order_Month
- Order_Month_Name
- Order_Day
- Year-Month

These features were used to analyse monthly and yearly sales trends.

5. Sales Trend Analysis

Monthly sales were calculated by grouping the transaction records by `Year_Month` and summing the `Sales` values. A line chart was created to visualize month-to-month changes in sales and identify periods of relatively high and low sales.

Yearly sales were calculated by grouping the data by `Order_Year` and summing the total `Sales`. A yearly sales trend chart was created to observe the overall direction of sales across the available years.

The monthly and yearly analysis provides an understanding of historical sales patterns and trends. These historical patterns were then used as the basis for developing the sales forecasting model.

6. Forecasting Methodology

For sales forecasting, the monthly sales dataset was used as the input for the forecasting model.

A `Time_Index` feature was created to represent the chronological order of each month. This numerical feature allows the Linear Regression model to learn the overall sales trend over time.

The dataset was divided chronologically into training and testing sets using an 80:20 split. The first 80% of the historical data was used to train the model, while the remaining 20% was used to evaluate its performance.

A Linear Regression model from Scikit-learn was trained using `Time_Index` as the independent variable and `Sales` as the dependent variable.

The trained model was then used to generate predictions for the testing period and forecast sales for the next six months.

A chronological split was used instead of a random split because sales forecasting is a time-based problem and the order of observations must be preserved.

7. Model Evaluation

The Linear Regression forecasting model was evaluated using three performance metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

- **MAE:** Measures the average absolute difference between actual and predicted sales.
- **MSE:** Measures the average squared difference between actual and predicted sales and gives more importance to larger errors.
- **RMSE:** Represents the prediction error in the same unit as the sales values.

Lower values for all three metrics indicate better forecasting performance.

The model performance was also visually evaluated by comparing actual sales with predicted sales for the testing period. This comparison helps determine how closely the model follows the actual sales pattern.

Model Evaluation Results:

- MAE: 18560.28591345771
- MSE: 557870823.4211354
- RMSE: 23619.289223453263

8. Future Sales Forecast

The trained Linear Regression model was used to forecast sales for the next six months beyond the historical dataset.

A future time index was generated for the upcoming six months, and the trained model was used to predict the corresponding sales values. The forecast results were visualized using a line chart to show the expected sales trend over the forecast period.

The six-month forecast provides an estimate of future sales based on the historical time-based trend captured by the Linear Regression model.

The forecast results can help businesses with:

- Sales planning
- Inventory management
- Resource allocation

- Revenue planning
- Short-term business decision-making

9. Business Insights

The sales analysis and forecasting results provide useful information for understanding historical performance and expected future sales.

Key business insights from the project include:

- Historical monthly and yearly sales analysis helps identify changes in sales performance over time.
- The six-month forecast provides an estimate of expected future sales based on the historical trend.
- Forecasted sales can support short-term sales and revenue planning.
- Businesses can use the forecast to plan inventory and resources more effectively.
- Identifying increasing or decreasing sales trends can help management make better operational decisions.
- The forecasting results should be used as a supporting tool rather than as the only basis for business decisions.

The analysis demonstrates how historical sales data can be transformed into useful business information through data analysis, visualization, and predictive modeling.

10. Limitations

Although the Sales Forecasting System provides useful estimates of future sales, the forecasting approach has some limitations:

- The project uses Linear Regression, which mainly captures the overall trend in sales over time.
- The model does not explicitly account for seasonal patterns, holidays, promotions, discounts, or special events.
- External factors such as economic conditions, market changes, and customer behavior are not included.
- The forecast is based only on historical sales data and assumes that the historical trend will continue in the future.
- More advanced time-series models may provide better forecasting performance when strong seasonal or cyclical patterns exist.

Therefore, the forecasts should be considered estimates and should be combined with business knowledge and other relevant information before making important decisions.

11. Conclusion

The Sales Forecasting System demonstrates how historical sales data can be analyzed and used to estimate future sales. The project involved data cleaning, preprocessing, monthly and yearly sales trend analysis, data visualization, and predictive modeling.

A Linear Regression model was developed using a time-based feature and evaluated using MAE, MSE, and RMSE. The model was then used to forecast sales for the next six months.

Overall, the project provides a practical example of how Python-based data analysis and machine learning techniques can support business forecasting and decision-making. Although the model has limitations, it establishes a foundation that can be improved using more advanced time-series forecasting techniques and additional business factors.